Counting Methods from the Perspective of Integer Partitions

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One Question

You're working as a cashier when a customer asks you to make change for a \$50 dollar bill.

How many possible combinations of bills could you hand back to them?

You can assume you have as many 1s, 2s, 5s, 10s, and 20s as you need.

Making change for \$50.

$$
\begin{aligned}
\$50 &= 20 + 20 + 10 \\
&= 20 + 20 + 5 + 5 \\
&= 20 + 20 + 5 + 2 + 2 + 1 \\
&= 20 + 20 + 5 + 2 + 1 + 1 + 1 \\
&= 20 + 20 + 5 + 1 + 1 + 1 + 1 + 1 \\
&= 20 + 20 + 2 + 2 + 2 + 2 + 2
\end{aligned}
$$

If we continue breaking \$2s, we get 5 more ways when we include two \$20 bills.

Partitions

Definition (Partition)

A partition of a non-negative integer, n, is a sequence of non-increasing natural numbers, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k$, such that

$$\sum_{i=1}^{k} \lambda_i = n.$$

Each λ_i is called a part of the partition.
The partition function, $p(n)$, counts the number of partitions of n.

An Example

Example (Partitions of 5)

$$
\begin{aligned}
5 &= 5 \\
&= 4+1 \\
&= 3+2 \\
&= 3+1+1 \\
&= 2+2+1 \\
&= 2+1+1+1 \\
&= 1+1+1+1+1
\end{aligned}
$$

$$p(5) = 7$$

- 1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77, 101, 135, 176, 231, 297, 385, 490, 627, 792, 1002, 1255, 1575, 1958, 2436, 3010, 3718, 4565, 5604,...

Generating Functions

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- $p(n)$ does not have an elementary formula.
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- Generating functions make our lives easier.
- The generating function for a sequence, a_n is a formal power series where the nth term in the sequence is the coefficient of the nth order term,

$$\sum_{n=0}^{\infty} a_n q^n.$$

Building a Generating Function for $p(n)$

- A generating function for $p(n)$ is then

$$\sum_{n=0}^{\infty} p(n)q^n.$$

- That doesn't tell us much about the value of $p(n)$.
- Notice that $(1+q+q^2)(1+q^2) = 1+q+2q^2+\cdots$
- $(1+q+q^2+q^3)(1+q^2)(1+q^3) = 1+q+2q^2+3q^3+\cdots$
- The coefficient of q is $p(1)$, q^2 has $p(2)$, and q^3 has $p(3)$.
- This isn't an accident!

Generating Function for Partitions

A generating function for $p(n)$ is

$$\begin{aligned}\sum_{n=0}^{\infty} p(n)q^n &= (1 + q^{(1)} + q^{(1+1)} + q^{(1+1+1)} + \cdots) \\ &\quad \times (1 + q^{(2)} + q^{(2+2)} + q^{(2+2+2)} + \cdots) \\ &\quad \times (1 + q^{(3)} + q^{(3+3)} + q^{(3+3+3)} + \cdots) \cdots \\ &= \frac{1}{(1-q)(1-q^2)(1-q^3)\cdots} \\ &= \prod_{k=1}^{\infty} \frac{1}{1-q^k}.\end{aligned}$$

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Let $(a;q)_\infty = \prod_{n=0}^{\infty}(1-aq^n)$ and $(a;q)_m = (a;q)_\infty/(aq^m;q)_\infty$ so that

$$\sum_{n=0}^{\infty} p(n)q^n = \frac{1}{(q;q)_\infty}.$$

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$$\sum_{n=0}^{\infty} p(n|\text{distinct parts })q^n = (1+q)(1+q^2)(1+q^3)\cdots = (-q;q)_\infty$$

- Let $p(n|\text{odd parts})$ be the number of partitions of n which have only odd parts.

$$\sum_{n=0}^{\infty} p(n|\text{odd parts })q^n = \frac{1}{(1-q)(1-q^3)(1-q^5)\cdots} = \frac{1}{(q;q^2)_\infty}$$

Surprising Fact #1

1. The number of partitions of n into odd parts is equal to the number of partitions of n into distinct parts.

$$(1+q)(1+q^2)(1+q^3)\cdots = \frac{1}{(1-q)(1-q^3)(1-q^5)\cdots}$$

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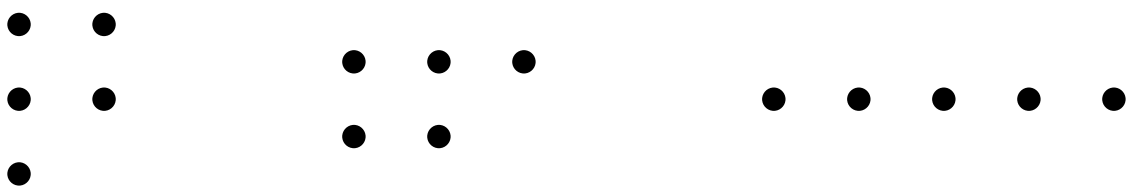
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$$(1+q)(1+q^2)(1+q^3)\cdots \times \frac{(1-q)(1-q^2)(1-q^3)\cdots}{(1-q)(1-q^2)(1-q^3)\cdots}$$
$$= \frac{(1-q^2)(1-q^4)(1-q^6)\cdots}{(1-q)(1-q^2)(1-q^3)\cdots}$$
$$= \frac{1}{(1-q)(1-q^3)(1-q^5)\cdots}$$

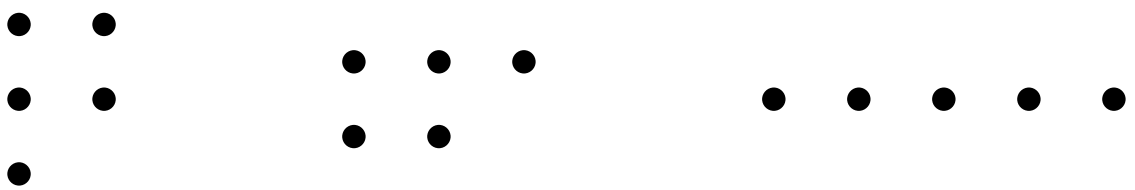
Ferrers digrams

Below from left to right are Ferrers diagrams of the partitions $2+2+1$, $3+2$, and 5.



Ferrers digrams

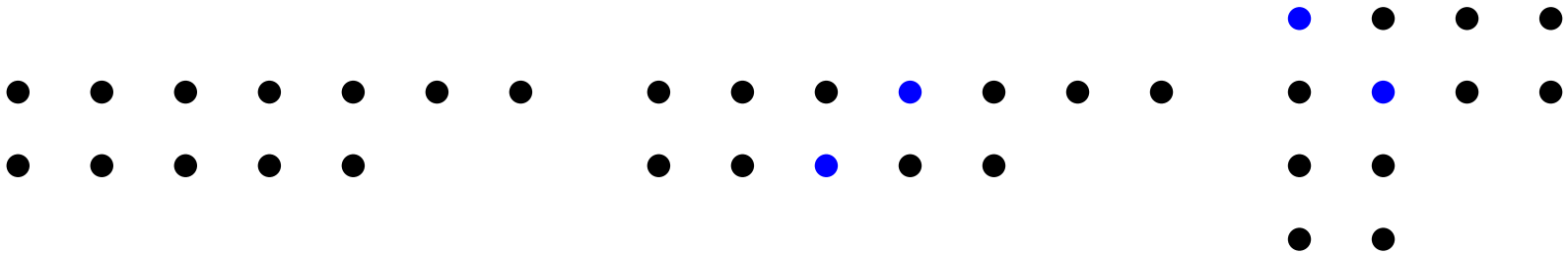
Below from left to right are Ferrers diagrams of the partitions $2+2+1$, $3+2$, and 5.



Ferrers diagrams are a visual representation of specific partitions, and they are a tool commonly used to develop intuition and prove results about partitions.

Surprising Fact #2

- The number of self-conjugate* partitions of n is equal to the number of partitions of n into distinct odd parts.
- How do we know? Consider the Ferrers Diagram for $7 + 5$



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- Recursions!

Developing a recursive formula for $p_k(n)$

- Call $p_k(n) = \#\text{partitions of } n \text{ into } k \text{ parts}$

$$p(n) = \sum_{k=0}^{n} p_k(n)$$

- If we can find a recursive formula for $p_k(n)$, then we can compute $p(n)$ from that!

Consequences of restricting by k

- Non-overlapping cases
- Control over (one) defining characteristic
- Write the Ferrers diagram for these three partitions:
  - $6+3+2$
  - $6+4+1$
  - $9+1+1$
- What happens when we remove 1 from each part?
  - $6+3+2 \rightarrow 5+2+1$
  - $6+4+1 \rightarrow 5+3$
  - $9+1+1 \rightarrow 8$

The Recursion

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The Recursion

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- Can we instead split into two cases based on whether or not there is a 1 in the partition?
  - $6+3+2 \to 5+2+1$
  - $6+4+1 \to 6+4$
  - $9+1+1 \to 9+1$

These insights lead to the following formula:

$$p_k(n) = p_{k-1}(n-1) + p_k(n-k)$$

- This recursion is good enough to compute $p(n)$ up to about $n = 10^5$, whereas the generating function in sage stops computing quickly at about $n = 100$.

Exact Formula for $p(n)$

- Did I lie a little at the beginning?

Exact Formula for $p(n)$

- Did I lie a little at the beginning? Maybe

Theorem (Rademacher [3])

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} \sqrt{k} \sum_{*} \omega(h,k) e^{\frac{-2\pi i n h}{k}} \frac{d}{dn} \left(\frac{\sinh\left(\frac{\pi}{k}\sqrt{\frac{2}{3}(n-\frac{1}{24})}\right)}{\sqrt{n-\frac{1}{24}}} \right)$$

where $* = 0 \leq h < k, ((h,k) = 1)$ *and*

$$\omega(h,k) = \exp\left(\pi i \sum_{r=1}^{k-1} \frac{r}{k}\left(\frac{hr}{k} - \left\lfloor \frac{hr}{k} \right\rfloor - \frac{1}{2}\right)\right)$$

Benefits of Rademacher's Expansion

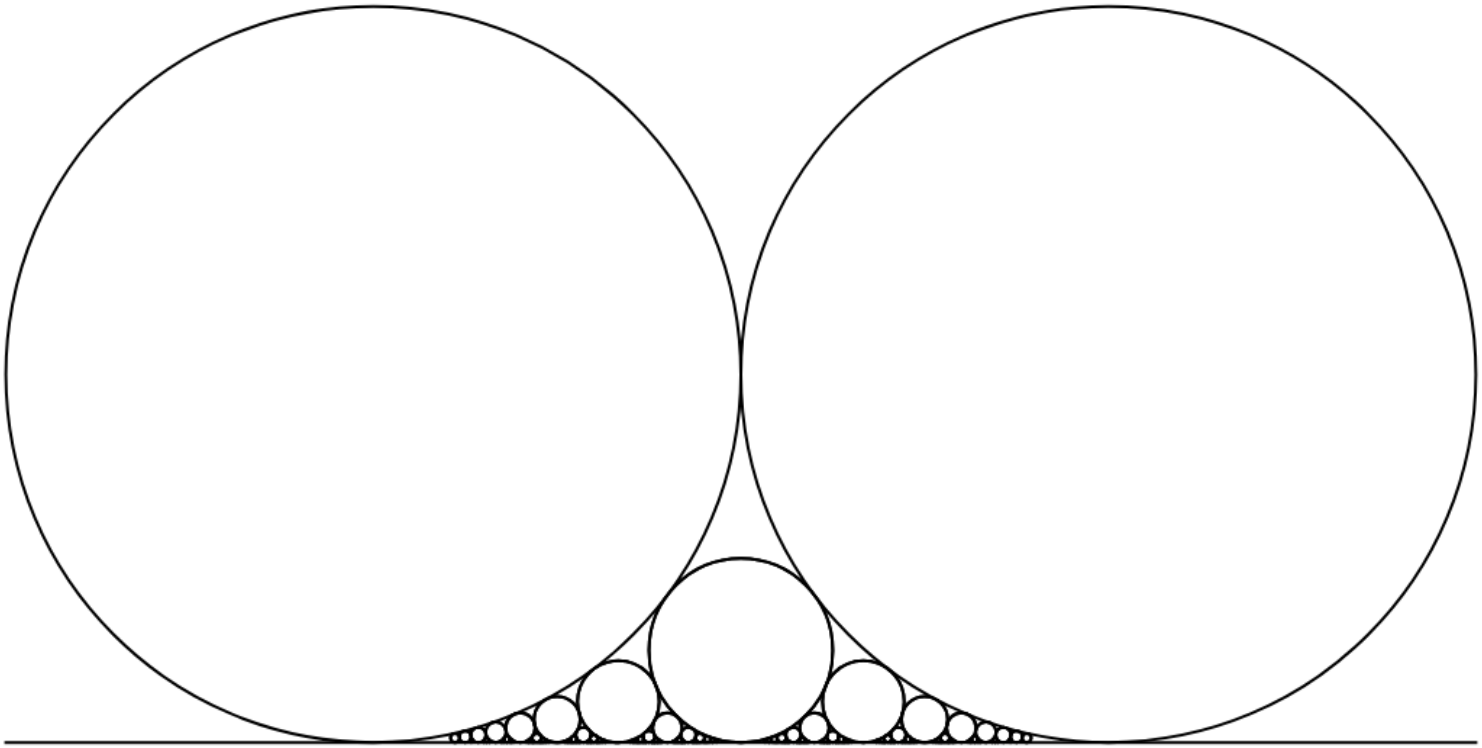
- The series converges very fast.
- The first term is the main term.

Corollary

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right)$$

- Asymptotic formulas are used to prove results "for large n."

A glimpse



[2]

One Necessary Answer

- There are 451 ways to make change for \$50 if we include the possibility of refusing and handing the \$50 bill back.
- It's the coefficient of the 50th term of

$$\frac{1}{(1-q^1)(1-q^2)(1-q^5)(1-q^{10})(1-q^{20})(1-q^{50})}.$$

- Looking for more details? Check out George Andrews's The Theory of Partitions[1]
- Thank you!

Bibliography I


George E Andrews.
The theory of partitions.
Number 2. Cambridge university press, 1998.


Jonathan Kogan and Ethan Zack.
Ford circles with variable radii. wolfram demonstrations project.


Hans Rademacher.
On the partition function p (n).
Proceedings of the London Mathematical Society, 2(1):241–254, 1938.